Downloaded from gsabulletin.gsapubs.org on October 21, 2010

Geological Society of America Bulletin

PALEOMAGNETIC SURVEY IN ARIZONA AND UTAH: PRELIMINARY RESULTS

S. K RUNCORN

Geological Society of America Bulletin 1956;67;301-316
doi: 10.1130/0016-7606(1956)67[301:PSIAAU]2.0.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geological Society of America Bulletin

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Copyright © 1956, The Geological Society of America, Inc. Copyright is not claimed on any material prepared by U.S. government employees within the scope of their employment.

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

BULLETIN OF THE GEOLOGICAL SOCIETY OF AMERICA
VOL. 67, PP. 301-316, 9 FIGS. MARCH 1956

PALEOMAGNETIC SURVEY IN ARIZONA AND UTAH: PRELIMINARY RESULTS

BY S. K. RUNCORN

ABSTRACT

Measurements of the remanent magnetization of sediments from the Colorado Plateau, ranging in age from the Algonkian to the Cretaceous, are reported. In some cases the stability of the magnetizations is clear, but in other cases instability is present to a degree for which corrections can be made. The original magnetizations are presumed to have been along the geomagnetic field at the time of deposition. Thus the position of the mean magnetic pole for five epochs is determined and compared with the pole positions calculated on the basis of measurements of strata in Great Britain. The agreement is sufficiently close to give strong support to the hypothesis of polar wandering.

CONTENTS

TEXT

	Page
Introduction	301
Acknowledgments	303
Collection and measurement of specimens	304
Magnetic stability	305
Summary of paleomagnetic directions	308
General statement	308
Precambrian formation	309
Upper Paleozoic formations	309
Mesozoic formations	311
Discussion of results	315
References cited	316

ILLUSTRATIONS

Figure	Page
1—Direction of magnetization of Hakati shale (Precambrian)	302
2—Direction of magnetization of Naco sandstone (Carboniferous-Pennsylvanian)	303
3—Directions of magnetization of Supai formation (Permian)	304
4—Direction of magnetization of Cutler formation (Permian)	305
5—Direction of magnetization of the Organ Rock tongue of the Cutler Formation (Permian)	309
6—Direction of magnetization of Moenkopi formation (Lower Triassic)	311
7—Direction of magnetization of Springdale sandstone (Triassic)	312
8—Direction of magnetization of Dakota sandstone (Cretaceous)	313
9—Migration of pole since Precambrian time	314

INTRODUCTION

Creer, Irving, and Runcorn (1954) and Runcorn (1955a) have suggested, mainly from consideration of paleomagnetic measurements in Great Britain, that the geographical pole has migrated slowly over the earth's surface throughout geological time. This study was based on rocks in which the natural remanent magnetization could be presumed to be the original magnetization. Thus rocks with considerable magnetic stability were used, and, where possible, the hypothesis that their magnetizations were acquired early in their history was tested. For such a study it is necessary to exclude rocks which have been subjected to circumstances that might appreciably change the magnetization, either through heating on deep burial or by diagenesis. Apart from random effects, it is reasonable to suppose that the original remanent magnetization of lavas and fine-grained red sandstones is in the same direction as that of the geomagnetic field at the time of cooling or deposition, respectively, when a correction is made for any subsequently acquired geological dip. Such rocks are found to have directions of magnetization, the mean of which in any one geological epoch has a direction characteristic of that epoch and in general different from ad-

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

jacent epochs. Within many such series of rocks, however, the direction shows frequent sharp reversals of sign. It is probable that this is accounted for by reversals of the polarity of the geomagnetic field. Between oppositely magnetized strata are often found thin zones of intermediate magnetization which presumably correspond to the relatively short periods during which the geomagnetic field is undergoing reversal of polarity.

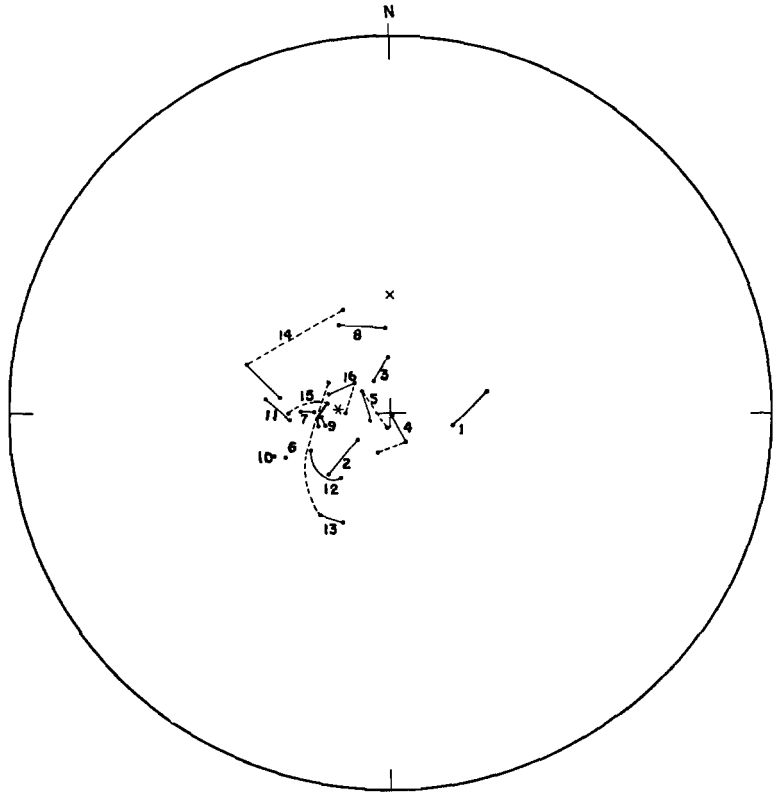


FIGURE 1.—DIRECTION OF MAGNETIZATION OF HAKATAI SHALE (PRECAMBRIAN)

Now the direction of magnetization of a single hand specimen is acquired usually within an interval *short* on the time scale of the geomagnetic secular change. The latter causes the field at any time to be substantially different from the mean dipole field with appreciable nondipole components. Thus a number of specimens, well spaced stratigraphically, will have directions of magnetizations with a random scatter not only from imperfect alignment of the magnetization along the field and from experimental errors but also from the effect of the geomagnetic secular variation.

But the mean direction should be that of the mean dipole of the epoch. In Tertiary time, as has clearly been shown by Hospers (1955), the mean field direction, inferred from measurements in Europe, is that of a geocentric dipole with its axis along the axis of the Earth's rotation. In pre-Tertiary times this is not so; therefore, unless relative movements of the continents are postulated, paleomagnetic results show that the mean dipole axis has varied with time. Creer, Irving, and Runcorn (1954) accordingly have calculated positions of the dipole axis, or mean magnetic poles, for different geological periods. It seems certain (Runcorn, 1955a) that these poles will be the same as the pole of rotation for the corresponding epoch.

A necessary stage in the test of this hy-

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

pothesis consists of the study of paleomagnetic directions throughout the geological column in some other area of the globe to examine how closely the positions of the pole inferred from these measurements agree with those inferred from the European measurements. Measurements of rocks from the Southwestern United States are particularly valuable for this purpose as this area lies about 90° along a great circle from Great Britain. In the summer of 1954 the writer collected specimens of various ages from the Grand Canyon and Colorado plateau, and the results of this preliminary survey are described in this paper.

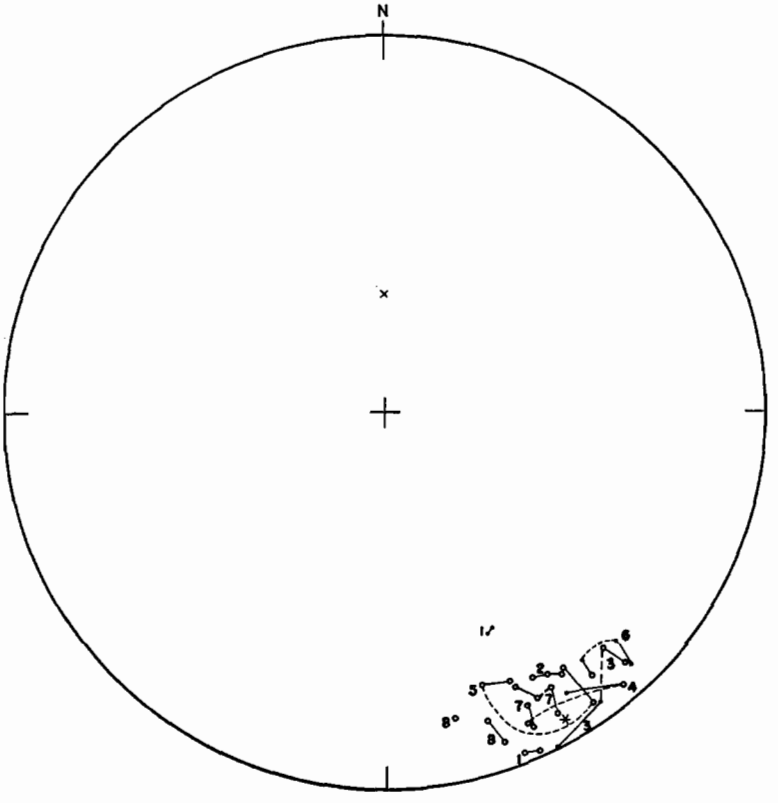


FIGURE 2.—DIRECTION OF MAGNETIZATION OF NACO SANDSTONE (CARBONIFEROUS-PENNSYLVANIAN)

ACKNOWLEDGMENTS

The author was enabled to visit the U.S.A. in the summer of 1954 by an invitation to participate in a Conference on Rock Magnetism held under the auspices of the National Science Foundation and the Institute of Geophysics of the University of California.

The collection of specimens was made possible by generous support from the Penrose Bequest of The Geological Society of America and by the invitation of the Director of the Museum of Northern Arizona, Flagstaff, to use the facilities there. The writer is grateful for much helpful advice from Dr. Henry R. Aldrich, from Mr. J. R. Balsley of the U. S. Geological Survey, and from Mr. L. F. Brady and Dr. J. F. Lance of the Museum of Northern Arizona. He is indebted to the Superintendents of the Grand Canyon and Zion and Bryce national parks for permission to collect specimens. Mr. R. R. Dorsey gave great help in the field work. Mr. J. C. Belshé made a preliminary compila tion of the results presented in this paper. The writer is especially grateful to Mr. G.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

Turnbull for carrying out most efficiently the measurements and much of the computation of the results described.

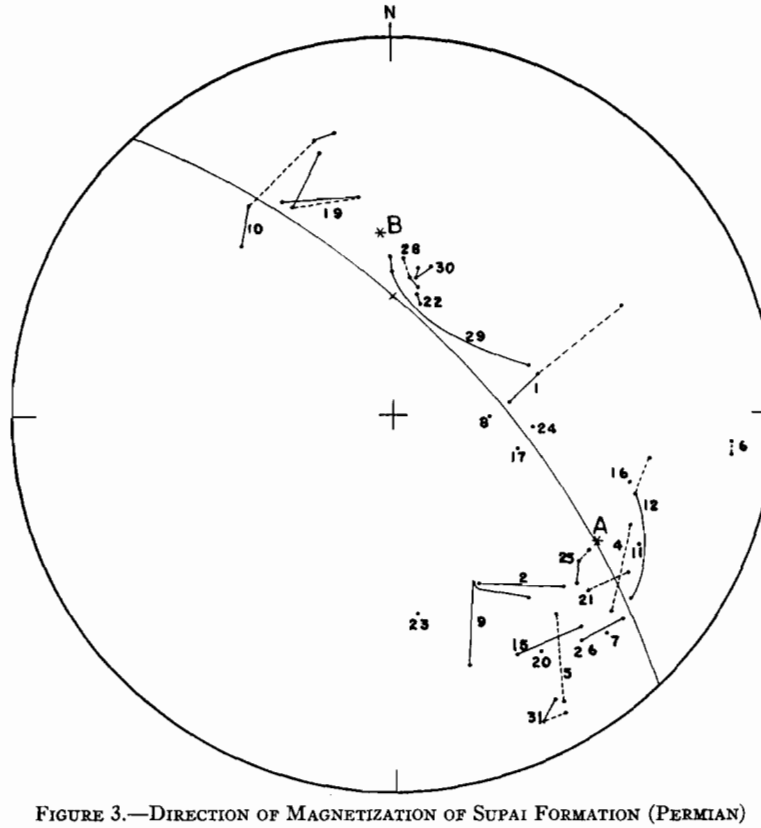


FIGURE 3.—DIRECTION OF MAGNETIZATION OF SUPAI FORMATION (PERMIAN)

COLLECTION AND MEASUREMENT OF SPECIMENS

Collection was limited to fine-grained red sandstones, siltstones, and shales. Previous work on the paleomagnetism of sediments has demonstrated (Runcorn, 1955a) that red sandstones are usually magnetized fairly strongly and have considerable magnetic stability, presumably because of the presence of hematite. Normally it is only fairly fine-grained well-bedded sandstones within one formation that are magnetized in a reasonably uniform direction. The settling of coarser material apparently occurs faster and in more turbulent water, so that mechanical forces have considerable influence in the orientation of the iron-oxide mineral grains. Sampling was limited to rocks whose original orientation can be determined. The present orientation was determined with a Brunton compass and marked on the upper surface as strike and dip marks. From each specimen one, or more commonly two, 3.5-cm diameter cores were cut with the axes perpendicular to the bedding plane. The cores were sliced into discs (usually two) about 1 cm thick.

The directions of magnetization of the discs were measured on the Cambridge astatic magnetometer following the procedure of Collinson, Creer, Irving, and Runcorn. The declination (relative to the present geographical north) and the inclination of the magnetic vector of the sample were measured relative to the original horizontal. The measurements are essentially observations of the angular deflections of a horizontal magnet suspended by a fine fiber when the rock specimen is raised beneath it. The high sensitivity required is obtained by removing the control of

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

the geomagnetic field on the suspended magnet. With the axis of the disc vertical and coincident with the magnet, the deflections obtained in various azimuths determine the declination.

nonuniformity of intensity of magnetization in addition to purely observational errors.

The instrument's limit of sensitivity through the period covering these measurements is

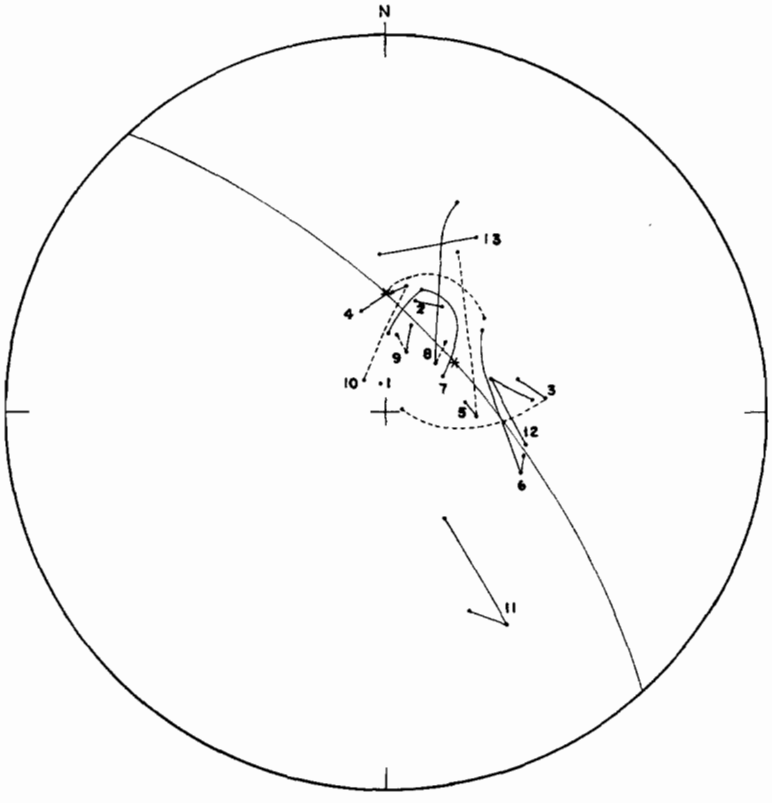


FIGURE 4.—DIRECTION OF MAGNETIZATION OF CUTLER FORMATION (PERMIAN)

In the present experiments the determination of the vertical component of magnetization was made by taking two series of measurements with the geometric center of the disc displaced 0.25 cm on one side of the vertical through the magnet for the first series of measurements of the deflection of the suspended system and the same distance on the other side for the second.

The observations yield more information than would be needed to determine the direction of magnetization on the assumption that this was adequately represented by a dipole at the center of the disc. Thus the reported directions of magnetizations are mean values calculated from a set of observations. This procedure therefore reduces errors arising from

estimated to be such as to give an average of 7.5×10^{-7} cgs e.m.u./cm^3 for the limit of detectable magnetization.

MAGNETIC STABILITY

The directions of magnetization of magnetically stable specimens from one formation will be scattered randomly, and this scatter will therefore nearly be eliminated in the mean direction. The directions may be represented by a number of unit vectors directed from a point or by their poles on the unit sphere. The mean direction is obtained by adding vectorially the number N of the unit vectors. The direc-

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 1.—DIRECTIONS OF MAGNETIZATION, HAKATAI SHALE

Specimen and Core No.	Distance along trail (Feet)	Disc 1		Disc 2	
		Declination	Inclination	Declination	Inclination
1	0	N 77 E	60.8	S 78.3 E	70.8
2	20	S 45.0 W	64.4	S 48. W	77.4
3	70	N 25 W	79.5	N 1.5 W	73.5
4-a	140	S 61 E	89.4	S 31 W	79.9
4-b		S 15 W	77.4		
5-a	180	S 8 W	85.4		
5-b		N 52 W	79.3	S 68 W	84.4
6	1000	S 67 W	57.0		
7	1090	N 89 W	67.6	N 89 W	63.7
8	2070	N 2 W	64.6	N 39 W	60.0
9	2220	S 79 W	70.6	S 86.5 W	70.0
10	2270	S 69 W	54.2		
11	2650	S 96 W	60.7	N 83.5 W	53.8
12	3520	S 64.6 W	64.3	S 37 W	66.1
13-a	3520	N 62 W	70.2		
13-b		S 34 W	54.2	S 23 W	55.1
14-a	3520	N 24 W	57.2		
14-b		N 83 W	57.9	N 71 W	47.3
15-a	3520	N 81 W	71.6	S 78.5 W	69.0
15-b		S 89 W	60.4		
16-a	3520	N 90 W	77.6		
16-b		N 48 W	77.0	N 73 W	71.3

TABLE 2.—DIRECTIONS OF MAGNETIZATION, NACO SANDSTONE

Specimen and Core No.	Disc 1		Disc 2		Disc 3	
	Declination	Inclination	Declination	Inclination	Declination	Inclination
1-a	S 22.4 E	−1.7	S 24.1 E	−0.9		
1-b	S 26.1 E	25.0	S 22.8 E	24.3		
2	S 28.9 E	−12.8	S 31.5 E	−12.1	S 33.8 E	−10.5
3-a	S 42.3 E	−9.8	S 43.8 E	−5.3		
3-b	S 36.6 E	2.9	S 26.9 E	0.2		
4-a	S 24.7 E	−5.8				
4-b	S 40.8 E	−3.0	S 32.5 E	7.3		
5-a	S 24.8 E	−14.3	S 19.4 E	−15.3		
5-b	S 35.6 E	−3.3	S 34.2 E	−11.5		
6-a	S 44.1 E	4.7	S 45.5 E	9.3		
6-b	S 38.1 E	10.2	S 37.9 E	−7.1		
7-a	S 29.8 E	−5.2	S 31.0 E	−9.6		
7-b	S 25.7 E	−8.7	S 24.7 E	−5.3		
7-b*	S 25.5 E	−12.9	S 29.7 E	−9.5		
8	S 18.1 E	−8.4	S 19.9 E	−4.1		
8*	S 12.7 E	−10.4				

* Repeat measurements after two months' interval

tion of this resultant vector (of length R) is that of the mean. About this mean pole on the unit sphere may be described the circle of confidence subtending the half angle α, calculated from N and R (Fisher, 1953). This circle encloses the true direction with a probability of 95 per cent. The dispersion K, a measure of the scatter, may also be calculated. The examples

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 3.—DIRECTIONS OF MAGNETIZATION, SUPAI FORMATION

Specimen and Core No.	Distance down Kabaib Trail	Disc 1		Disc 2		Disc 3	
		Declination	Inclination	Declination	Inclination	Declination	Inclination
1-a	200	N 74.5 E	47.1	N 84.2 E	55.9		
1-b		N 64.4 E	22.8				
2	280	S 44.0 E	25.1	S 26.1 E	37.2		
4-a	770	S 47.0 E	14.9				
4-b				S 64.2 E	20.7		
5-a	940	S 29.8 E	7.3				
5-b		S 38.6 E	21.8				
6-a	1370	S 82.7 E	6.7				
6-b		S 84.9 E	6.9				
7	1800	S 43.5 E	12.7				
8	2000	S 88.2 E	61.2				
9-a	2150	S 36.0 E	28.0	S 25.0 E	37.6	S 16.4 E	20.4
10-a	2620	N 41.8 W	28.5	N 34.4 W	22.7		
10-b		N 11.3 W	15.5	N 15.4 W	16.0		
11	2720	S 61.8 E	17.9				
12-a	2990	S 51.4 E	13.8	S 71.3 E	22.3		
12-b		S 79.8 E	21.0				
15-a	4000	S 37.8 E	16.6	S 26.5 E	19.1		
16-a	4520	S 73.6 E	24.0				
17	4960	S 74.6 E	52.5				
19-a	5800	N 25.4 W	27.6	N 15.0 W	18.8		
19-b		N 8.6 W	29.4	N 2.7 W	25.1		
20	6000	S 31.3 E	17.3				
21-a	6500	S 55.5 E	16.8				
21-b		S 47.0 E	20.6				
22	6990	N 14.2 E	56.4	N 11.3 E	54.7		
23		S 6.3 E	34.0				
24-a		S 84.7 E	49.9				
25-a		S 46.4 E	23.6	S 51.5 E	26.1		
25-b		S 54.2 E	25.6				
26-a		S 47.1 E	12.1	S 39.2 E	14.8		
28-a		N 11.4 E	52.2	N 7.5 E	50.2		
28-b		N 3.8 E	45.4				
29		N 0.8 W	44.8	N 9.0	48.6	N 7.0 E	48.5
30		N 14.3 E	46.3	N 9.4 E	49.9	N 10.1 E	47.8
31-a	S	S 29.4 E	6.0				
31-b		S 25.4 E	6.1	S 29.0 E	8.2		

The following specimens are omitted because they possessed magnetizations too small for accurate measurements: 3, 4-a, 4-b, 7, 8, 9-b, 13, 14, 15-b, 16-b, 17, 18, 24-b, 27-a, 27-b.

of Figures 1 and 2 show cases in which this simple analysis of the results is appropriate.

In Figures 3, 4, and 7, however, the magnetizations in one series of rocks are distributed near a plane rather than about a point, and clearly a mean direction is unlikely to have a simple significance. At ordinary temperatures, through exposure to fields of the order of that of the earth, rocks can acquire an isothermal remanent magnetization (I.R.M.) in addition to their original stable magnetization. The theory of I.R.M. has been worked out by Néel (1949; 1955); the intensity rises as the logarithm of the time of exposure to the field. Although such I.R.M. may, in certain cases, obscure the stable magnetization after quite short exposures to the field, in general a substantially long build-up time is necessary before I.R.M. becomes of comparable importance to the original magnetization. Thus

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

the I.R.M. acquired by a rock in nature will be directed parallel to the field of a geocentric dipole along the present geographical axis, which we will call the present axial dipole field. Thus it may be expected that the directions of the magnetizations of specimens from one series of rocks, neglecting scatter, will lie near a plane containing this latter direction and the stable magnetization. The directions will lie between these limits clustering toward the one or other depending on the relative importance of the I.R.M. in different samples. Thus in such cases the directions as plotted on a stereographic projection should be scattered along a great circle passing through the mean direction of all the specimens and the present axial dipole field direction at that locality. It is with this principle that we proceed to interpret the measured directions of magnetization where the calculation of a simple mean is clearly inappropriate.

TABLE 4.—DIRECTIONS OF MAGNETIZATION, CUTLER FORMATION

Specimen and Core No.	Disc 1		Disc 2		Disc 3	
	Declination	Inclination	Declination	Inclination	Declination	Inclination
1	N 9.7 W	81.7	(Not measured)			
2-a	(Not measured)		N 71.7 E	54.5		
2-b	N 15.3 E	56.1	N 28.1 E	55.0		
3-a	N 78.3 E	84.8				
3-b	N 84.7 E	44.1	N 75.5 E	50.7		
4-a	N 13.4 W	59.9	N 1.9 E	55.3		
4-b	N 46.4 E	50.7				
5-a	N 82.6 E	66.0	S 86.3 E	62.8		
5-b	N 24.6 E	40.2				
6	N 50.0 E	53.2	S 65.6 E	4.8	S 72.0 E	48.2
7	N 57.9 E	70.0	N 16.7 E	53.4	N 2.7 E	66.8
8-a	N 19.1 E	29.6	N 45.7 E	69.3	N 40.2 E	62.3
8-b	N 40.2 E	62.3				
9-a	N 8.4 E	67.1				
9-b	N 19.2 E	71.2	N 16.3 E	63.1		
10-a	N 3.2 E	54.6	N 9.7 E	53.4		
10-b	N 34.7 W	78.8				
11	S 28.5 E	54.7	S 29.7 E	24.1	S 22.4 E	30.6
12	N 85.2 E	47.8	S 72.0 E	57.5	S 76.5 E	48.1
13	N 1.9 W	45.3	N 27.6 E	35.4		

TABLE 5.—DIRECTIONS OF MAGNETIZATION, ORGAN ROCK TONGUE OF CUTLER FORMATION

Specimen and Core No.	Disc 1		Disc 2	
	Declination	Inclination	Declination	Inclination
65-a	S 63.5 E	66.6	S 83.1 E	59.7
65-b			S 19.3 E	73.4
66-a	N 79.5 E	53.9	N 70.0 E	57.5
66-b			N 41.0 E	69.2

A single specimen may have a direction of magnetization quite different from the rest of its group. Such will be ignored in this discussion. Quite apart from the chance of mistakes in the process of orientation in the field or laboratory, very local disturbed conditions during or after deposition, which upset the process of magnetization, are always possible.

SUMMARY OF PALEOMAGNETIC DIRECTIONS

General Statement

The stereographic projection is used throughout this paper to plot the directions of magnetizations. Solid dots represent projections on the lower hemisphere—*i.e.*, positive inclinations (north-seeking pole downward); open circles represent the upper hemisphere—*i.e.*, negative inclinations (north-seeking pole upward). Directions of magnetization of discs

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

from the same core are joined by solid lines, measurements on cores from the same rock specimen by broken lines. The small cross represents the direction of the present axial dipole field in Arizona. Stars represent mean directions of all, or some, of the measurements on each formation.

In Figure 1 the directions are shown: Specimens 12–16 were collected a few feet above each other, but their scatter is no less than the scatter of the whole group.

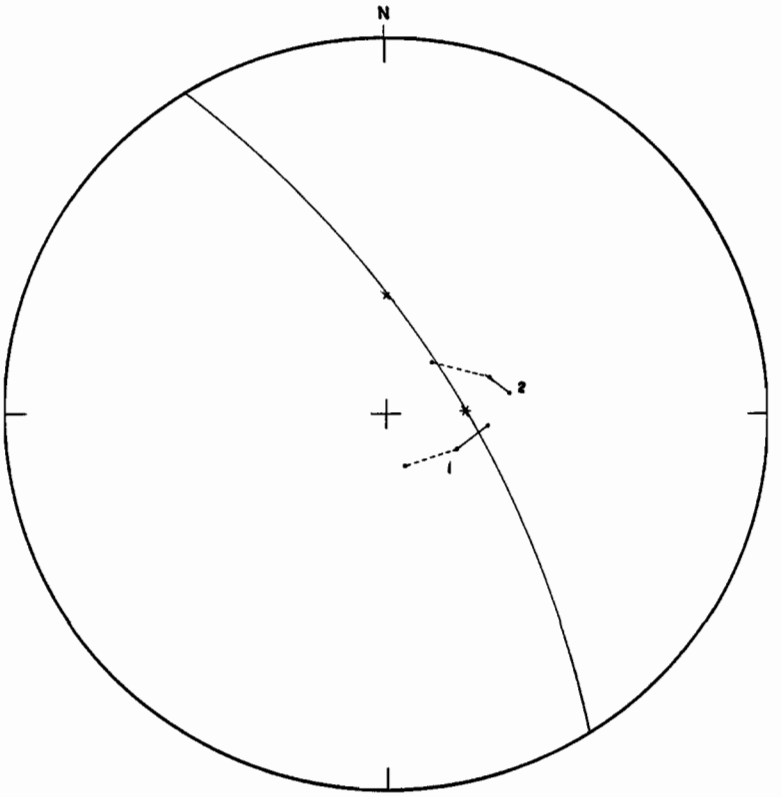


FIGURE 5.—DIRECTION OF MAGNETIZATION OF THE ORGAN ROCK TONGUE OF THE CUTLER FORMATION (PERMIAN)

Precambrian Formation

HAKATAI SHALE (Algonkian or late Precambrian): Sixteen specimens of the Hakatai shale were collected along the Kaibab trail in the Grand Canyon (Table 1). The first specimen was collected 264 feet farther down the Trail, from where the Kaibab fault crosses it. Omitting specimen 1 as being spurious, the statistics were calculated to be as follows:

Mean declination S. 87.7° W. } denoted by * on
Mean inclination +72.8° } Fig. 1
$N = 34$ $R =$ 32.475 $K = 21.6$
Circle of confidence $\alpha =$ 5.4°

Upper Paleozoic Formations

NACO SANDSTONE (Carboniferous-Pennsylvanian): Eight specimens (1–8) of the Naco sandstone spanning a thickness of about 200 feet were collected a quarter of a mile north of Carizzo Creek Bridge, Arizona, on the west bank on a road cut. Figure 2 and Table 2 show the direction of magnetization, and the following statistics were calculated:

Mean declination S. 30.3° E. } denoted by * on
Mean inclination −3.4° } Fig. 2
$N = 31$ $R = 30.25$ $K = 40.0$
Circle of confidence $\alpha =$ 4.1°

SUPAI FORMATION (Permian): Twenty-two specimens (1–22) were collected from the Supai formation in the Grand Canyon along

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 6.—DIRECTIONS OF MAGNETIZATION, MOENKOPI FORMATION

Specimen and Core No.	Disc 1		Disc 2		Disc 3	
	Declination	Inclination	Declination	Inclination	Declination	Inclination
1-a	N 35.3 W	29.3	N 44 W	21.1		
1-b	N 40.7 W	17.3				
2-a	N 44.7 W	27.7				
2-b	N 39.6 W	22.3	N 50.8 W	23.1		
3-a	N 10.9 W	31.9	N 2.8 W	39.2		
3-b	N 26.3 W	47.8				
4-a	N 24.4 W	41.4	N 13.3 E	55.2		
4-b	N 25.9 W	43.6				
5-a	S 3.0 W	57.2	S 4.0 W	55.0		
5-b	S 4.0 E	49.8				
6-a	N 12.3 W	33.9	N 8.4 W	39.4		
6-b	N 9.7 W	40.8				
7-a	N 22.5 E	24.7				
7-b	N 50.3 E	44.9				
8-a	N 4.7 W	47.3				
8-b	N 54.7 E	48.6				
9-a	N 31.2 E	22.6	N 31.0 E	16.3		
9-b	N 30.0 E	20.6				
10-a	S 55.3 E	18.0	S 36.4 E	28.5		
10-b	S 46.3 E	19.8				
11-a	N 32.4 E	21.0	N 37.7 E	31.8		
11-b	N 51.8 E	18.0	N 54.2 E	25.4		
12-a	N 29.2 W	11.0				
12-b	N 17.4 W	14.5	N 15.9 W	22.1		
13-a	N 2.0 W	9.4	N 2.8 W	11.7		
13-b	N 2.8 W	17.3				
14-a	N 9.1 W	11.3	N 14.3 W	11.1		
14-b	N 4.4 W	14.9	N 5.7 W	14.0		
15			N 32.9 E	−1.9	N 33.4 E	5.2
16-a	N 15.3 W	19.3	N 13.8 E	18.6		
16-b	N 4.3 E	34.5	N 14.5 E	37.6		
17	N 6.1 E	42.6	N 16.3 E	32.5	N 0. E	26.6
18-a	N 8.4 W	7.7	N 2.3 E	7.9		
18-b	N 4.3 E	9.8	N 1.7 E	7.0		

the Kaibab Trail; 5 specimens (23–27) were collected in a road cut south of Midgley Bridge, on U. S. Highway 88, 32.1 miles south of the Research center of the Museum of Northern Arizona, Flagstaff; 3 (28–30) from a quarry near Flagstaff; and 1 (31) from a quarter of a mile above Pendley's Bridge. Table 3 and Figure 3 show the results. In Table 3 the position at which the samples were collected is shown as measured from the 1.5 mile marker down the Kaibab Trail. The main group of specimens have southeast declinations. Omitting specimens 10, 19, 22, 28, 29, and 30, the following statistics were calculated:

Mean declination S. 47.3° E.
Mean inclination +22.9°
} denoted by the point A in the quadrant in Fig. 3.

$N = 34$ $R = 31.9$ $K = 15.7$
Circle of confidence $\alpha = 7.5°$

A small group (22, 28, 29, 30) center on the present direction of the earth's field, and another group (1, 8, 17, 24) lies near the great circle between this direction and the prevalent direction, shown on Figure 3. Clearly the former group is completely unstable, and the latter partially unstable. Considering specimens 10, 19, 22, 29, and 30 the following mean

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

direction (denoted by the point *B* toward the north of Fig. 3) was found: declination N. 4.0° W., inclination + 39.7°. The point *A* 6 miles south of Gouldings on the west side of the road 30–40 feet below the de Chelly formation. Table 5 and Figure 5 give the results.

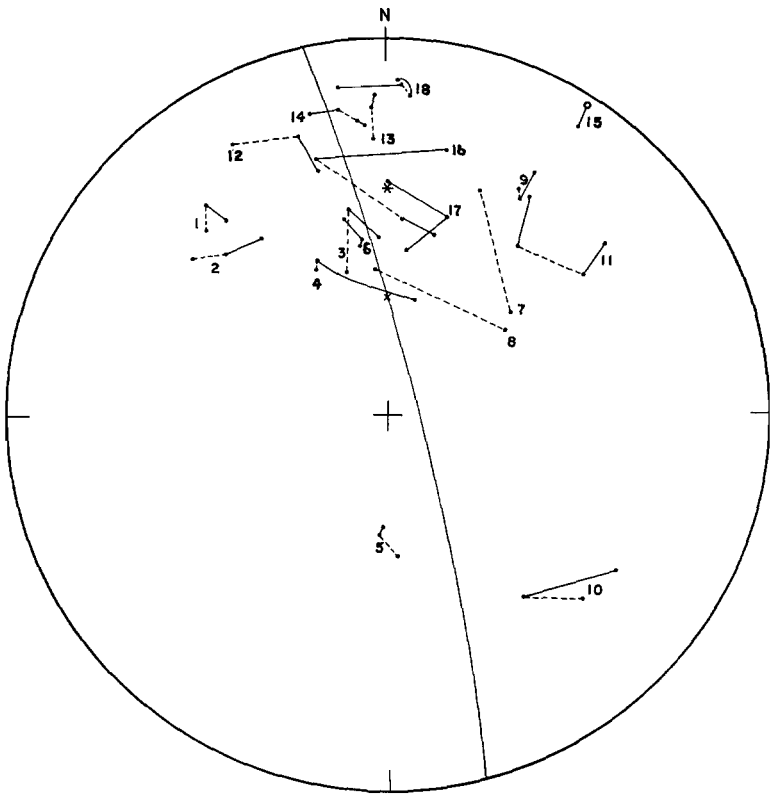


FIGURE 6.—DIRECTION OF MAGNETIZATION OF MOENKOPI FORMATION (LOWER TRIASSIC)

will be regarded as the paleomagnetic direction, although it is likely to be slightly in error, the true one lying farther southeast along the great circle.

CUTLER FORMATION (Permian): Thirteen specimens (1–13) were collected from the Cutler formation in Monument Valley spanning a thickness of about 100 feet. Figure 4 and Table 4 show the results. The magnetizations lie along a great circle through the present dipole field direction and their mean direction. This great circle is nearly the same as that for the Supai formation in Figure 3. These rocks are therefore inferred to be partially unstable. The mean declination is N. 54.5° E., and mean inclination is +65.0°.

ORGAN ROCK TONGUE OF THE CUTLER FORMATION (Permian): Two specimens were collected Again the great circle through the mean value and the present dipole field is similar to those of Figures 3 and 4, and we infer that these samples are partially unstable.

These three Permian formations show varying degrees of instability, but the great circles are in good agreement. This is consistent with their having the same original magnetization. Consequently the mean value *A* of the magnetization of the Supai is best taken to be the paleomagnetic direction for Permian time.

Mesozoic Formations

MOENKOPI (Lower Triassic): Two specimens (1, 2) were collected 60 feet below the Shinarump formation in Zion National Park, just west of Springerville, Utah; 4 (3–6) from a road

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

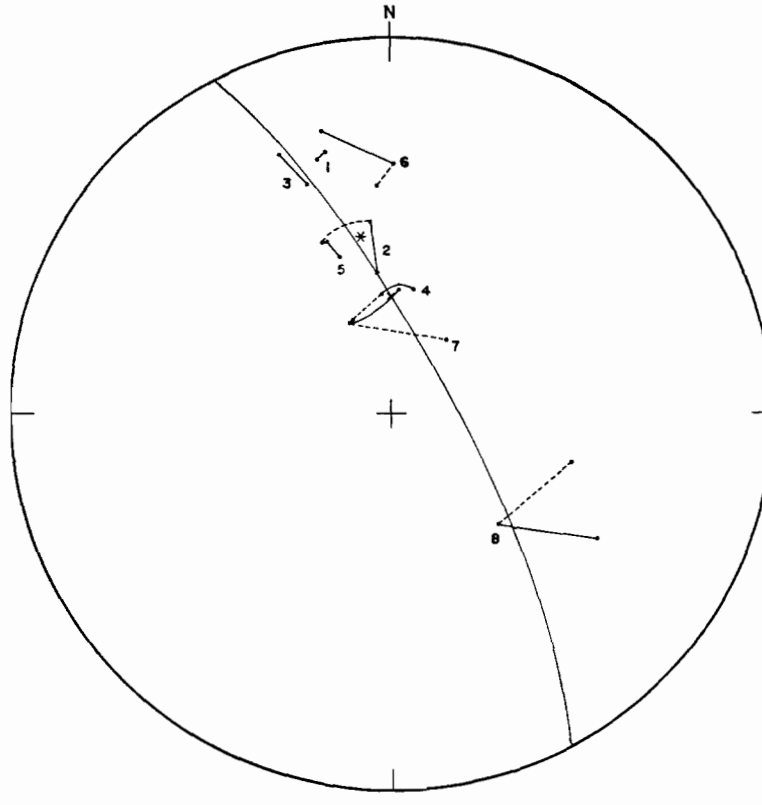


FIGURE 7.—DIRECTION OF MAGNETIZATION OF SPRINGDALE SANDSTONE (TRIASSIC)

TABLE 7.—DIRECTIONS OF MAGNETIZATION, SPRINGDALE SANDSTONE

Specimen and Core No.	Disc 1		Disc 2	
	Declination	Inclination	Declination	Inclination
1	N 13.8 W	18.4	N 15.6 W	19.6
2-c	N 21.3 W	37.1		
2-b	N 6.2 W	35.0	N 5.2 W	48.3
3	N 19.6 W	24.0	N 22.9 W	16.5
4-a	N 10. E	52.5	N 4.3 W	54.3
4-b	N 21.5 W	59.3		
5	N 17.9 W	42.3	N 20.2 W	37.1
6-a	N 13.4 W	14.8	N 0.9 E	22.5
6-b	N 3.1 W	27.2		
8-a	N 36.1 E	62.7		
8-b	N 23.3 W	59.9	N 3.4 E	53.1
7-a	S 58.9 E	25.8	S 43.8 E	46.6
7-b	S 74.8 E	38.2		

cut a mile from the east city limit sign in Flagstaff; 8 (7–14) from a road cut 2.8–3.8 miles south of Cameron; and 4 (15–18) 5 miles from Holbrook on the east bank of a road cut. Table 6 and Figure 6 give the results. Omitting specimens 5 and 10, the mean declination is N. 0.2° E., and the mean inclination is +27.4°, denoted by the star on Figure 6. The scatter is somewhat great. This may be connected with the fact that these rocks are often nonuniform in composition. Inclusions of gypsum are common. If specimens 1, 2, 9, 11, and 15 are omitted because they lie away from the main cluster of points and specimens 7, 8, and 16 are omitted because their internal magnetization seems unusually scattered, the great circle would lie as shown rather than through the mean point.

SPRINGDALE SANDSTONE (Triassic): Eight specimens (1–8) were collected in Zion National Park. Figure 7 and Table 7 show the results. Their directions lie near the great circle which has been drawn on Figure 7 through the mean direction. It may therefore be inferred that the

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 8.—DIRECTIONS OF MAGNETIZATION, DAKOTA SANDSTONE

Specimen and Core No.	Disc 1		Disc 2		Disc 3	
	Declination	Inclination	Declination	Inclination	Declination	Inclination
1-a	S 2.3 W	−65.5	S 2.2 W	−67.5		
1-a*	S 0.8 E	−68.7				
1-b	S 9.6 W	−65.3	S 5.7 W	−63.7		
2-a	S 35.6 E	−50.6				
2-b	S 24.3 E	−54.1	S 32.8 E	−51.1		
3-a	S 35.2 E	−62.7				
3-b	S 24.7 E	−60.4				
4-a	N 15.5 W	+61.8	N 30.4 W	+64.1		
4-b	N 7.5 W	+63.5	N 9.3 W	+65.8		
4-a*	N 4.6 E	+58.9	N 4.7 E	+62.8		
5-a	N 10.5 E	+39.1	N 15.5 E	+37.8		
5-b	N 2.7 E	+35.9	N 0.3 W	+36.7		
6	S 49.7 E	+7.9	S 47.8 E	−4.3	S 39.0 E	11.9

* Repeat measurements after two months' interval

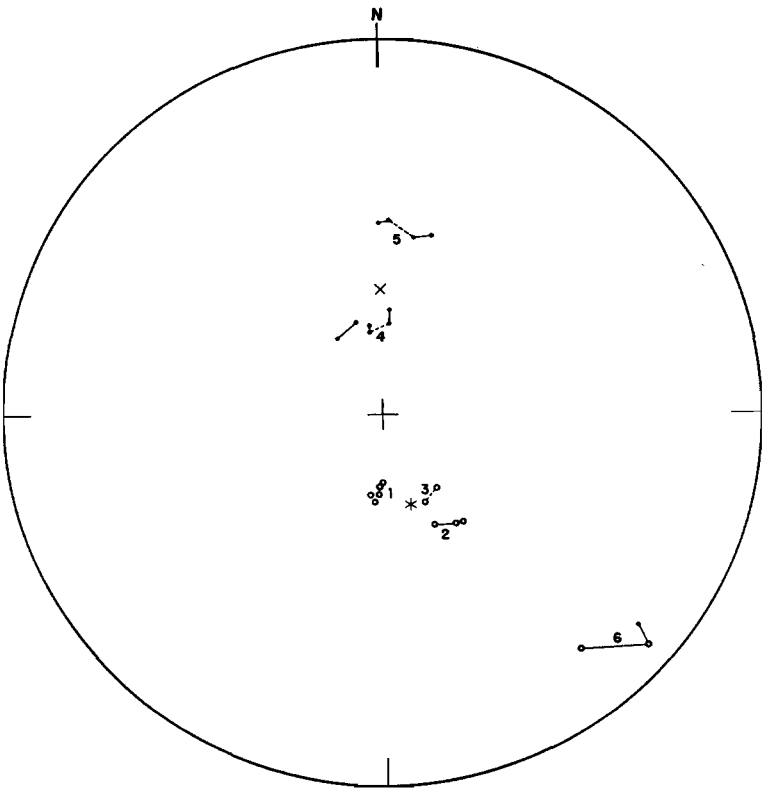


FIGURE 8.—DIRECTION OF MAGNETIZATION OF DAKOTA SANDSTONE (CRETACEOUS)

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

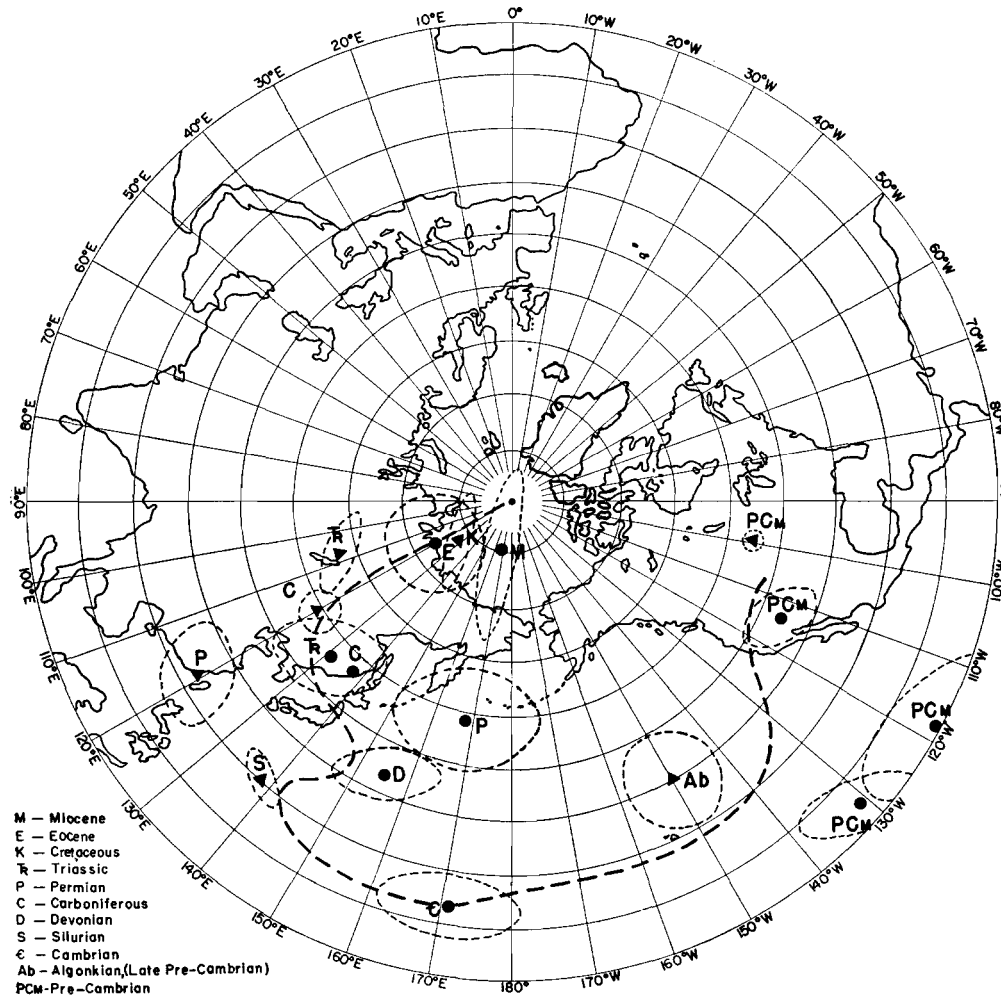


FIGURE 9.—MIGRATION OF POLE SINCE PRECAMBRIAN TIME
Based on paleomagnetic measurements in Europe and North America

rock is partially unstable. If 8 is omitted from the statistics we obtain

Mean declination N. 9.75° W. } shown by star
Mean inclination +38.8° } on Fig. 7
$N = 18$ $R = 17.0$ $K = 17$
Circle of confidence $\alpha =$ 8.7°

These two Triassic formations do not therefore give unambiguous results, but in comparing them it seems reasonable to believe that the true paleomagnetic direction for this period has a smaller dip than that of the present field and a declination just west of north. Therefore a point on the great circle on Figure 7 farther north than the mean point is taken as more probably the true one rather than the mean value of the measurements quoted above. A mean declination of N. 22° W. and a mean inclination of +16° are taken, but the possibility of considerable error must be borne in mind.

DAKOTA SANDSTONES (Cretaceous): Six specimens were collected, 1–3, 3 miles west of Showlow on the south of the U. S. Highway and 4–6, 4 miles south of Showlow on the west bank of a road cut. The results are shown in Figure 8 and Table 8. There is no obvious evidence

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

that specimens 4 and 5 are stable, and but 1, 2, 3 clearly are. Whatever interpretation is put on this reversed magnetization, these results show that the Cretaceous pole cannot have been far from the present one. Specimens 1, 2, and 3 give a mean declination of S. 16° E. and a mean inclination of −62°.

TABLE 9.—DIRECTIONS OF MAGNETIZATION OF ROCKS FROM ARIZONA AND UTAH AND INFERRED POLE POSITIONS

Age	Formation	Declination	Inclination	Magnetic Lat.	Sign of field	Position of North Pole Lat.	Long.
Cretaceous	Dakota sandstones	S 16°E	−62°	43°N	−	76.5°N	127°E
Triassic	Springdale and Moenkopi	N 22°W	+16°		+	54.9°N	107°E
Permian	Supai	S 47.3°E	+22.9°	11.9°N	+	26.0°N	119°E
Carboniferous-Pennsylvanian	Maco sandstones	S 30.3°E	−3.4°	1.7°S	+	48.7°N	120°E
Pre-Cambrian	Hakatai shales	S 87.7°W	+72.8°	58.2°N	−	30.7°N	148°W

TABLE 10.—SEMI-AXES OF OVALS OF CONFIDENCE OF POLE POSITIONS

	β	α
EUROPEAN ROCKS		
Precambrian (Lower Torridonian)	8°	4.6°
Precambrian (Longmyndian)	13.2°	6.8°
Precambrian (Upper Torridonian)	7.8°	5.1°
Cambrian	9.6°	5.8°
Devonian	10.0°	5.0°
Triassic	13.0°	7.0°
Tertiary Eocene	12.1°	9.1°
Tertiary Miocene	17.6°	15.5°
AMERICAN ROCKS		
Precambrian (Michigan)	2°	2°
Algonkian	9.7°	8.7°
Silurian	3.6°	2.2°
Pennsylvanian	4.1°	4.1°
Permian	8.0°	6.4°
Triassic	8.7°	4.3°
Cretaceous	11.0°	8.5°

DISCUSSION OF RESULTS

From the true paleomagnetic directions for each of the geological epochs, the direction of the mean magnetic, and therefore the geographical, axis for that epoch may be calculated. It is not possible, because of the reversals of polarity, to distinguish between the north and the south geographical poles. However, by assuming that the axis of rotation travels slowly on a fairly smooth path, Creer, Irving, and Runcorn (1954) have traced the motion of the north geographical pole backward in time from its present position, and they find it seems not to have migrated outside the present northern hemisphere since Precambrian time. It is convenient to call the field of a geomagnetic dipole oriented with its south pole pointing to the geographical north (or the pole star on the celestial sphere) negative and a field with opposite polarity positive. Thus the present geomagnetic field has a negative sign. It is then possible with the above assumption about the motion of the axis of rotation to determine the sign of the field corresponding to each paleomagnetic direction. Irving (1954, Thesis, Cambridge Univ.) finds frequent reversals in the Torridonian (upper Precambrian) sandstones of northwest Scotland, as do Clegg, Almond, and Stubbs (1954) in the Keuper marls (Triassic) of England. However, reversals in Paleozoic time seem rather rare; Creer (1955, Thesis, Cambridge Univ.) finds from somewhat extensive sampling, positive fields in Cambrian through Devonian time and again in Permian time. Quite possibly in certain periods reversals of the geomagnetic field were very frequent and in other periods quite rare. The results described in this paper show that positive fields existed in Pennsylvanian and Permian time and negative

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

fields in Algonkian and Triassic time, though in no cases is the sampling extensive enough to prove that fields of the other polarity did not occur in these epochs.

In Figure 9 the position of the north geographical pole for various epochs, inferred by Creer, Irving, and Runcorn (1954) from European data, is plotted by solid circles on the aximuthal equidistant projection of the northern hemisphere. The triangles denote the pole positions obtained by measurements of North American rocks including those from Arizona and two, the Silurian and mid-American Precambrian poles, deduced by Creer (1955, Thesis, Cambridge Univ.) from some measurements by Graham. The pole in the late Precambrian, inferred from the results on the upper Torridonian (Aultbea and Applecross groups) and Longmyndian formations or from the Hakatai shales, was in the east Pacific in low latitudes. Earlier in Precambrian time it had been in Arizona as inferred from the direction of magnetization of the Diabaig group of the Torridonian. In early Paleozoic time the pole was in the western Pacific near the tropics. In Late Paleozoic and Mesozoic time it was near the Asiatic mainland in higher latitudes.

The positions of the poles shown are surrounded by closed curves computed from the circles of confidence of the respective formations so that the true pole lies within these curves with a probability of 95 per cent. The semi-axes of these curves, α and β, along the great circle joining the pole position and the locality and the great circle orthogonal to it, respectively, are given in Table 10. This estimate of accuracy will be misleading if few samples were taken close to each other; the Permian pole deduced from British measurements and that of the Silurian and Precambrian measurements of Graham are likely to have artificially high accuracies for this reason.

Runcorn (1955b) considered that the agreement between the European and American results is sufficiently good to conclude that no appreciable continental drift of the amount postulated by Wegener and du Toit—*i.e.*, 60° of longitude—could have taken place, at least since Precambrian time. Only the Permian poles are displaced by this amount, and this is probably the result of inadequate sampling. Consequently if the coincidence of the mean magnetic axis with the geographical axis is accepted, these measurements must be regarded as giving considerable support to the reality of polar wandering.

References Cited

Clegg, J. A., Almond, M., and Stubbs, P. H. S., 1954, The remanent magnetization of some sedimentary rocks in Britain: Philos. Mag., v. 45, p. 583–598

Creer, K. M., Irving, E., and Runcorn, S. K., 1954, The direction of the geomagnetic field in remote epochs in Great Britain: Jour. Geomag. and Geoelect., v. 6, p. 163–168

Fisher, R. A., 1953, Dispersion on a sphere: Royal Soc. A Pr., v. 217, p. 295–305

Hospers, J., 1955, Rock magnetism and polar wandering: Jour. Geology, v. 63, p. 59–74

Néel, L., 1949, Theorie du trainage magnetique des ferromagnetiques en grains fins avec applications aux terras cuites: Ann. Geophysics, v. 5, p. 99–136

—— 1955, Some theoretical aspects of rock magnetism: Adv. in Physics, v. 4, p. 191–243

Runcorn, S. K., 1955a, Rock magnetism—geophysical aspects: Adv. in Physics, v. 4, p. 244–291

—— 1955b, Palaeomagnetism of sediments from the Colorado Plateau: Nature, v. 176, p. 505

Museum of Northern Arizona, Flagstaff, Ariz., and Dept. of Geodesy and Geophysics, Cambridge University, Cambridge, England

Manuscript Received by the Secretary of the Society, June 24, 1955

Project Grant 659–54